Project Initialization and Planning Phase

Date	6 July 2026
Team ID	xxxxxx
Project Name	Credit Card Fraud Detection System using Machine Learning
Maximum Marks	2 Marks

Ideation:

Ideation helps a team explore multiple possible approaches before committing to a solution. For our Credit Card Fraud Detection System, our team evaluated different detection approaches and selected the one best suited to deliver accurate, real-time fraud predictions.

Possible Solutions

Approach	Description	Limitation
Rule-based detection	Flags transactions using fixed, manually defined rules such as amount thresholds or location checks.	Rigid and fails to catch new or evolving fraud patterns.
Statistical analysis	Detects anomalies by analyzing transaction patterns and deviations from normal behavior.	Struggles with complex, non-linear relationships between features.
Machine learning	Learns fraud patterns directly from historical transaction data to classify new transactions.	Requires quality labeled data and careful model evaluation.

Chosen Solution

Machine Learning

Our team selected machine learning as the core approach, training and comparing four algorithms: Logistic Regression, Decision Tree, Random Forest, and XGBoost, to identify the model that best balances accuracy and reliability for fraud classification.

Reasons for Selection

- Machine learning models can automatically learn complex fraud patterns from historical data, unlike fixed rules.
- Comparing multiple algorithms allows the team to choose the model with the best accuracy and generalization.
- Random Forest and XGBoost handle non-linear relationships and feature interactions effectively.
- The trained model can generate real-time predictions through a simple web interface.
- This approach scales well as more transaction data becomes available over time.

Conclusion

The ideation phase helped our team compare rule-based, statistical, and machine learning approaches for fraud detection. Machine learning was selected as the most effective solution, and this decision guided the model development and evaluation stages that followed.